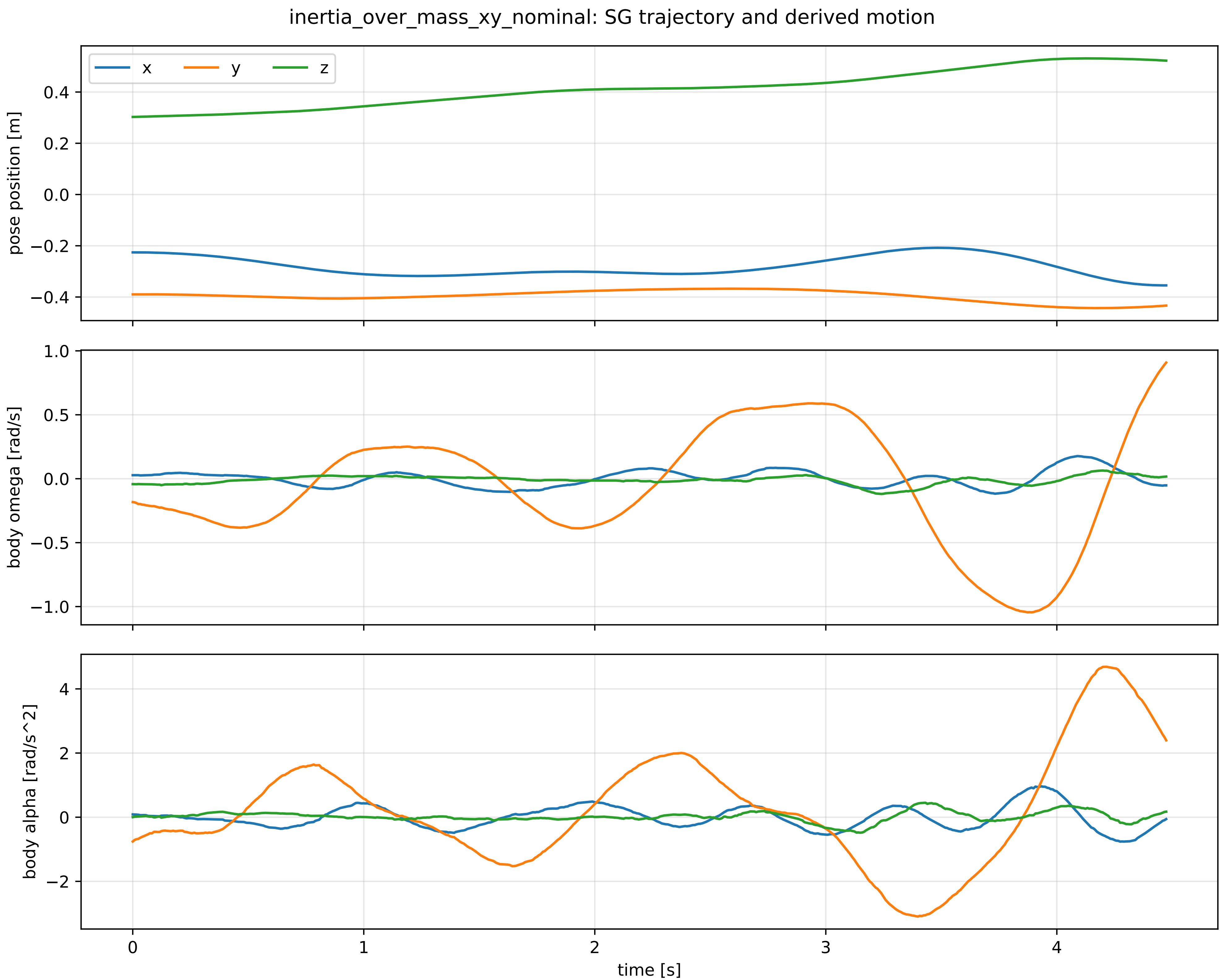
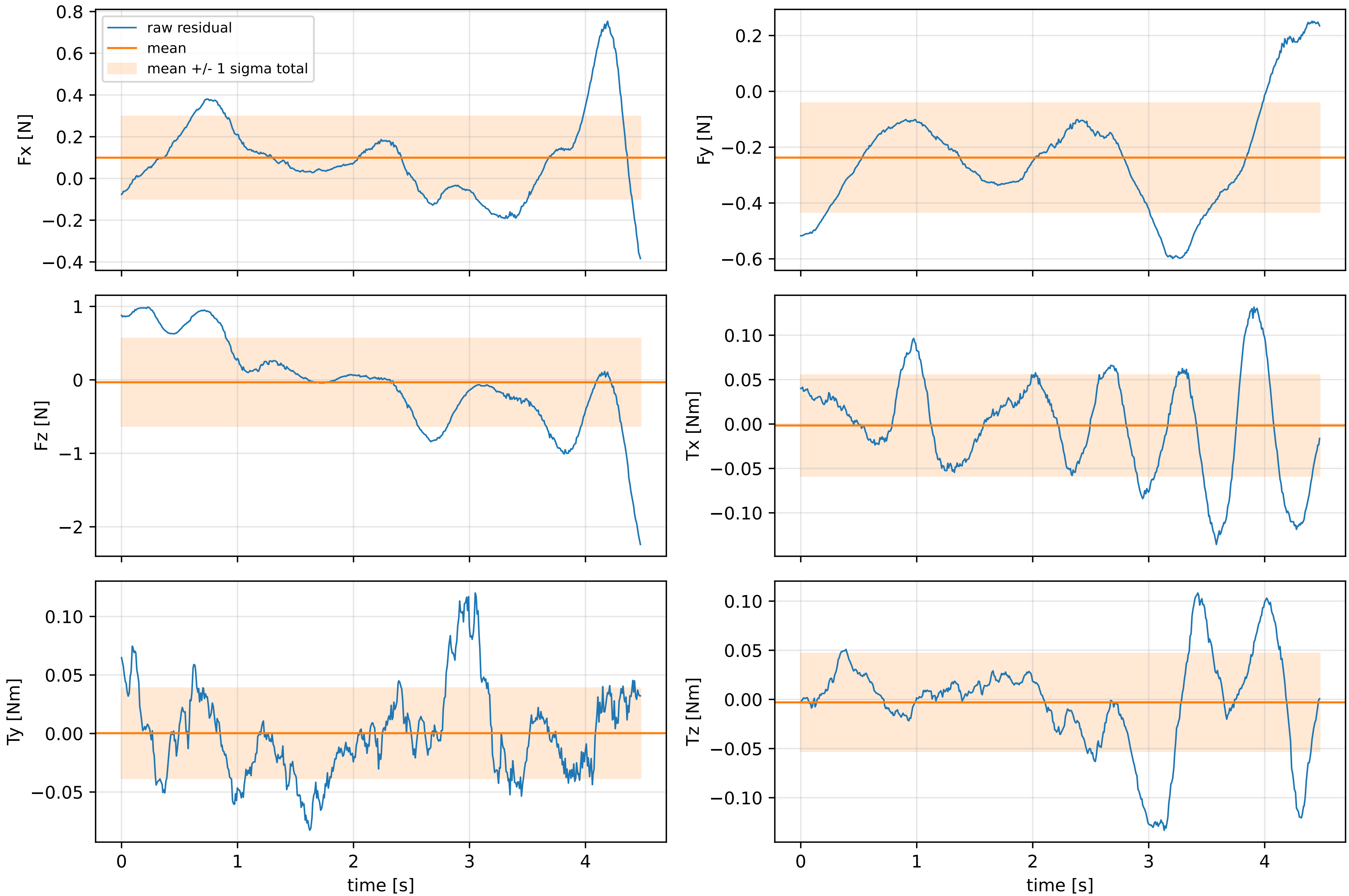
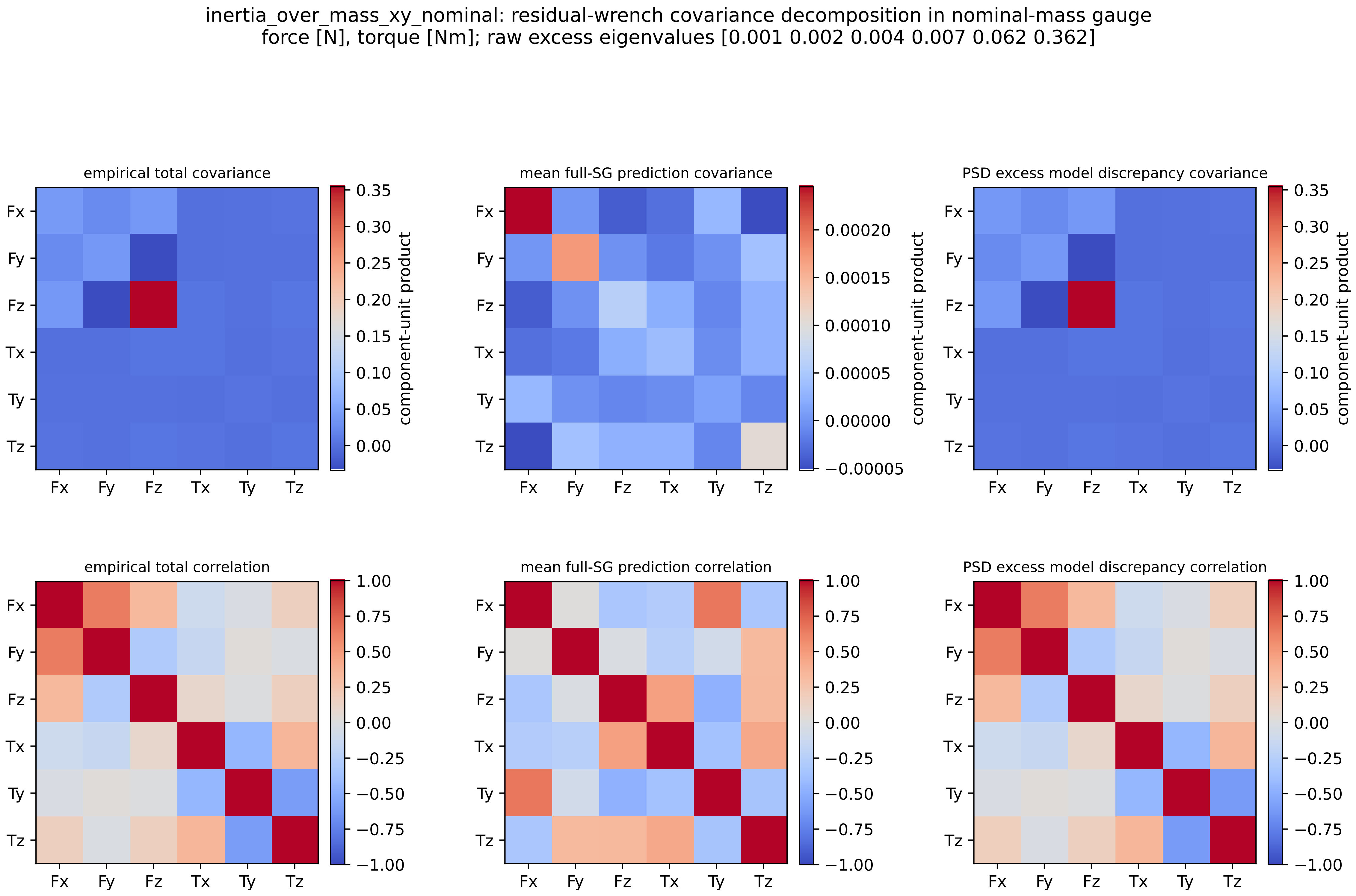
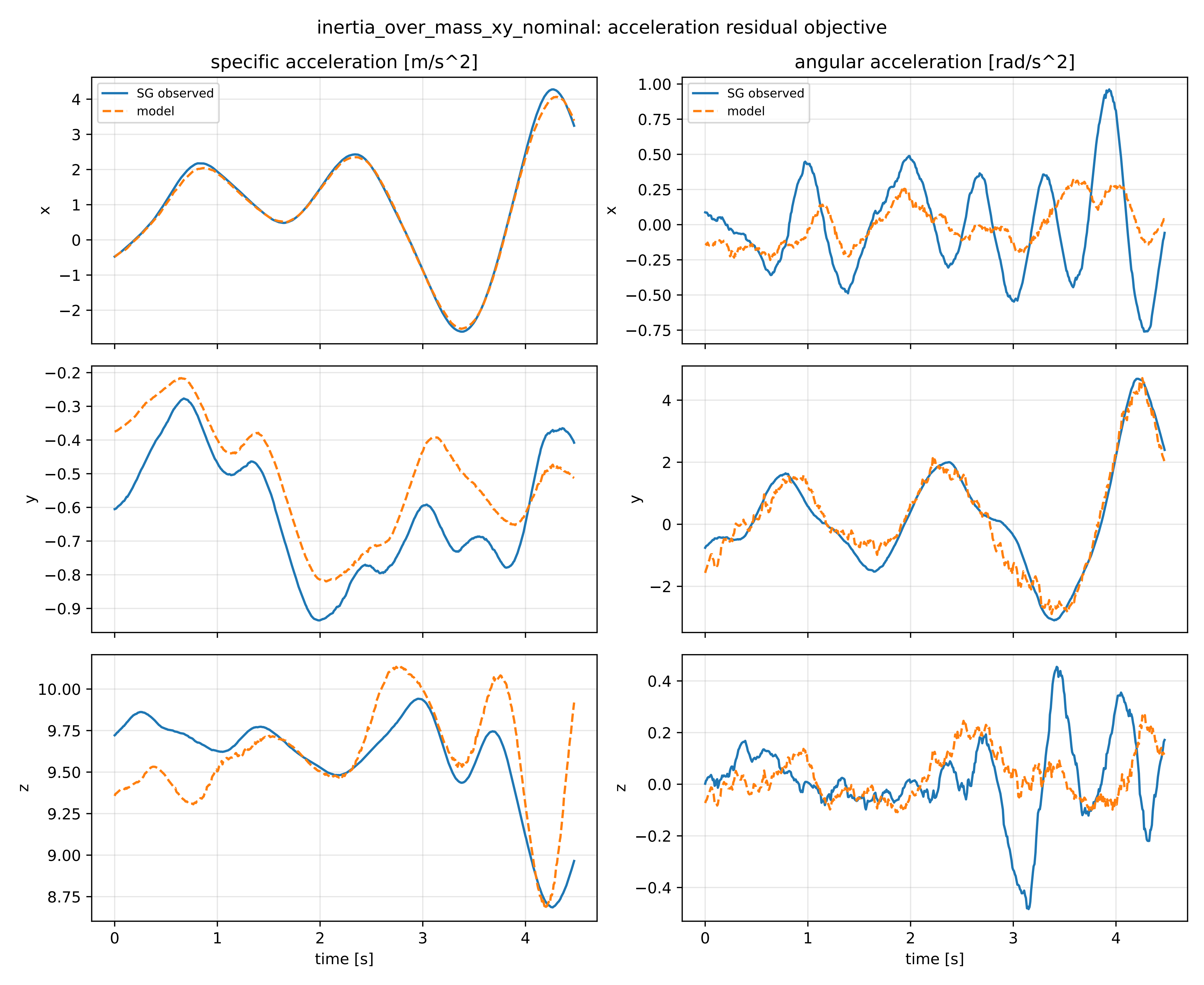

inertia_over_mass_xy_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]
0.4
0.2
0.0
−0.2
−0.4
1.0
0.5
0.0
−0.5
−1.0
4
2
0
−2
0
1
2
3
4

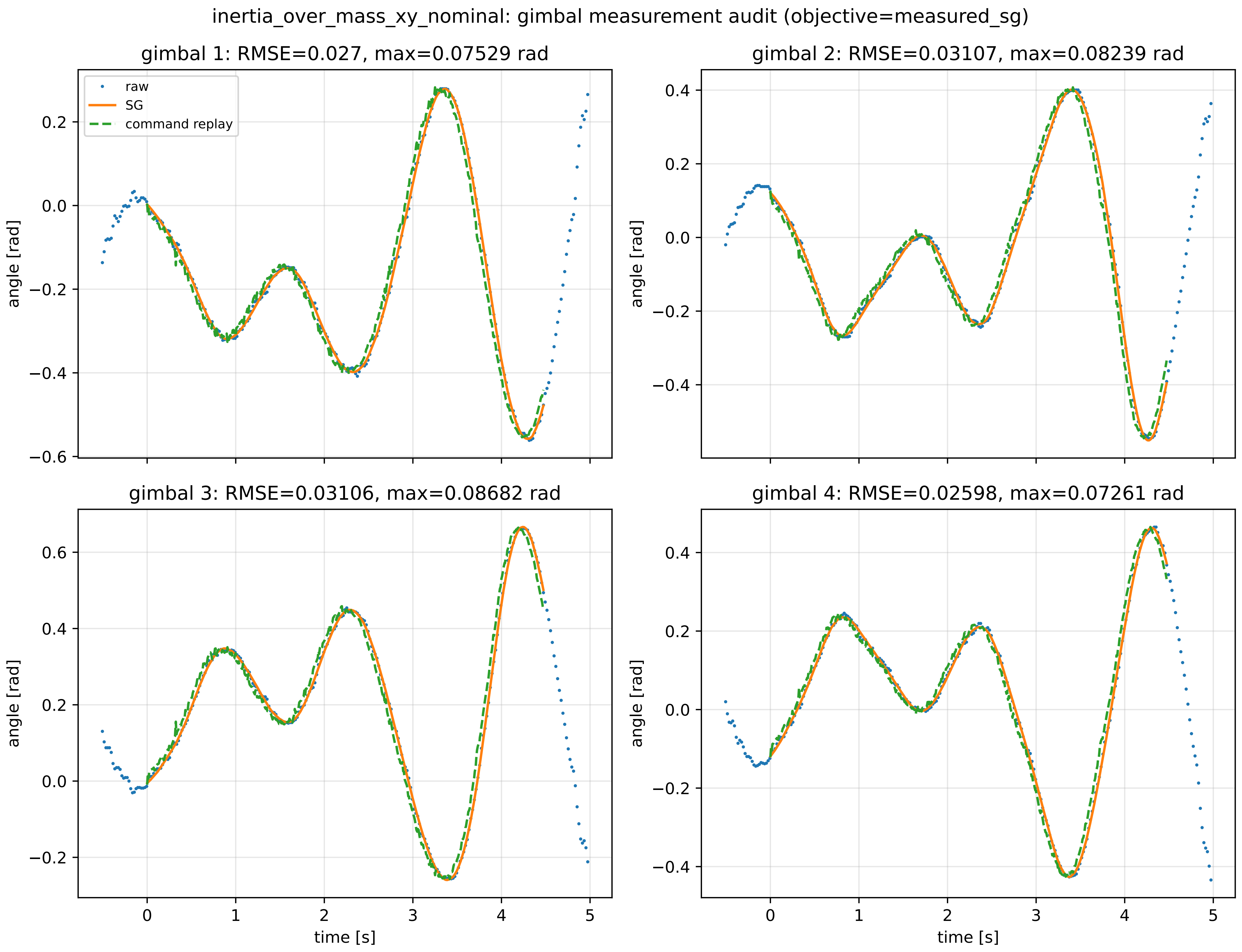
inertia_over_mass_xy_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



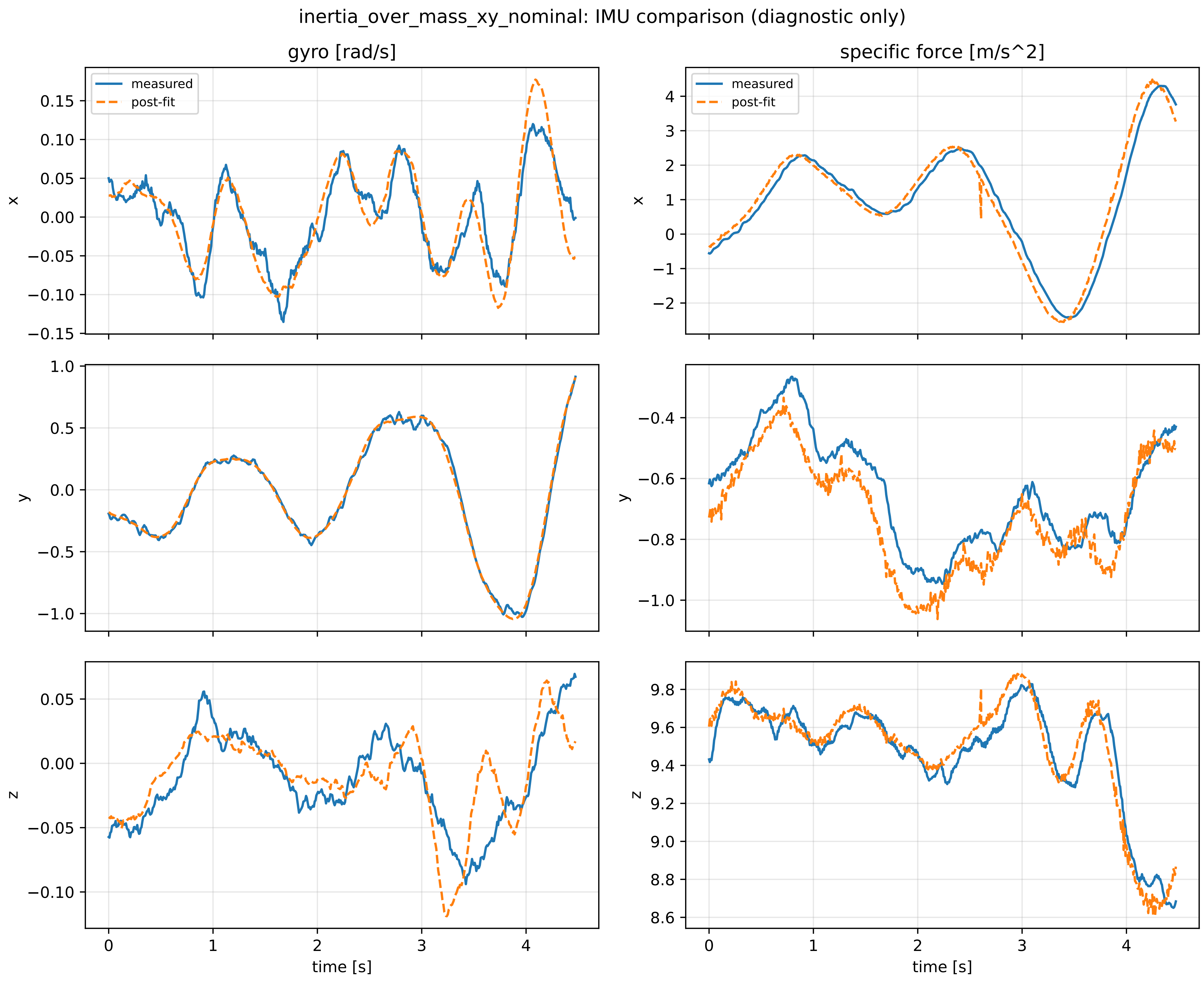
vector RMS about zero: force 0.7055 N, torque 0.084586 Nm; component std about mean: force [0.1971 0.1948 0.5954], torque [0.0567 0.0384 0.0497]


inertia_over_mass_xy_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.004 0.007 0.062 0.362]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
component-unit product
Fx Fy Fz Tx Ty Tz
0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00
0.00020 0.00015 0.00010 0.00005 0.00000 −0.00005
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00

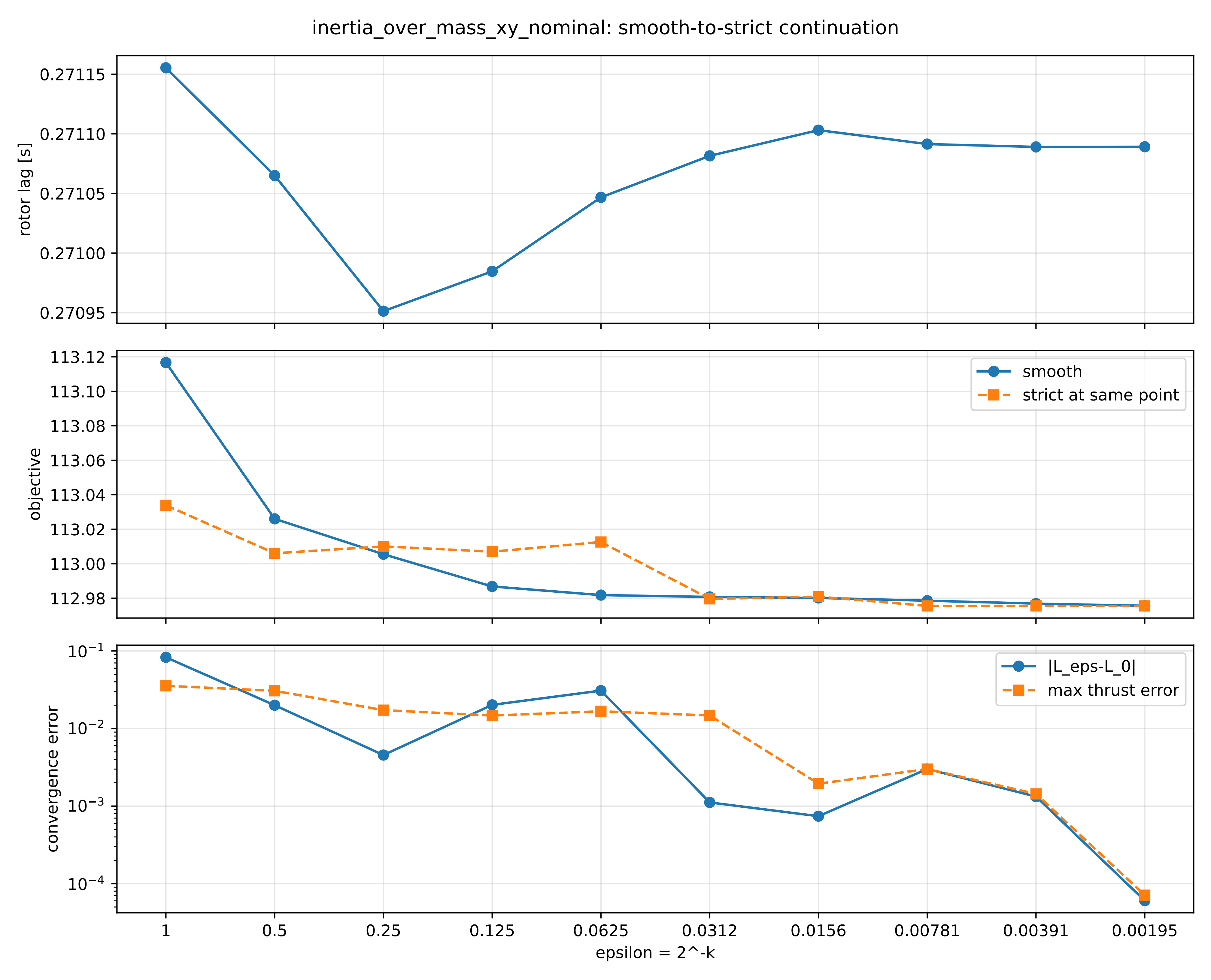

inertia_over_mass_xy_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



inertia_over_mass_xy_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.027, max=0.07529 rad
gimbal 2: RMSE=0.03107, max=0.08239 rad
gimbal 3: RMSE=0.03106, max=0.08682 rad
gimbal 4: RMSE=0.02598, max=0.07261 rad
raw
SG
command replay
angle [rad]
time [s]

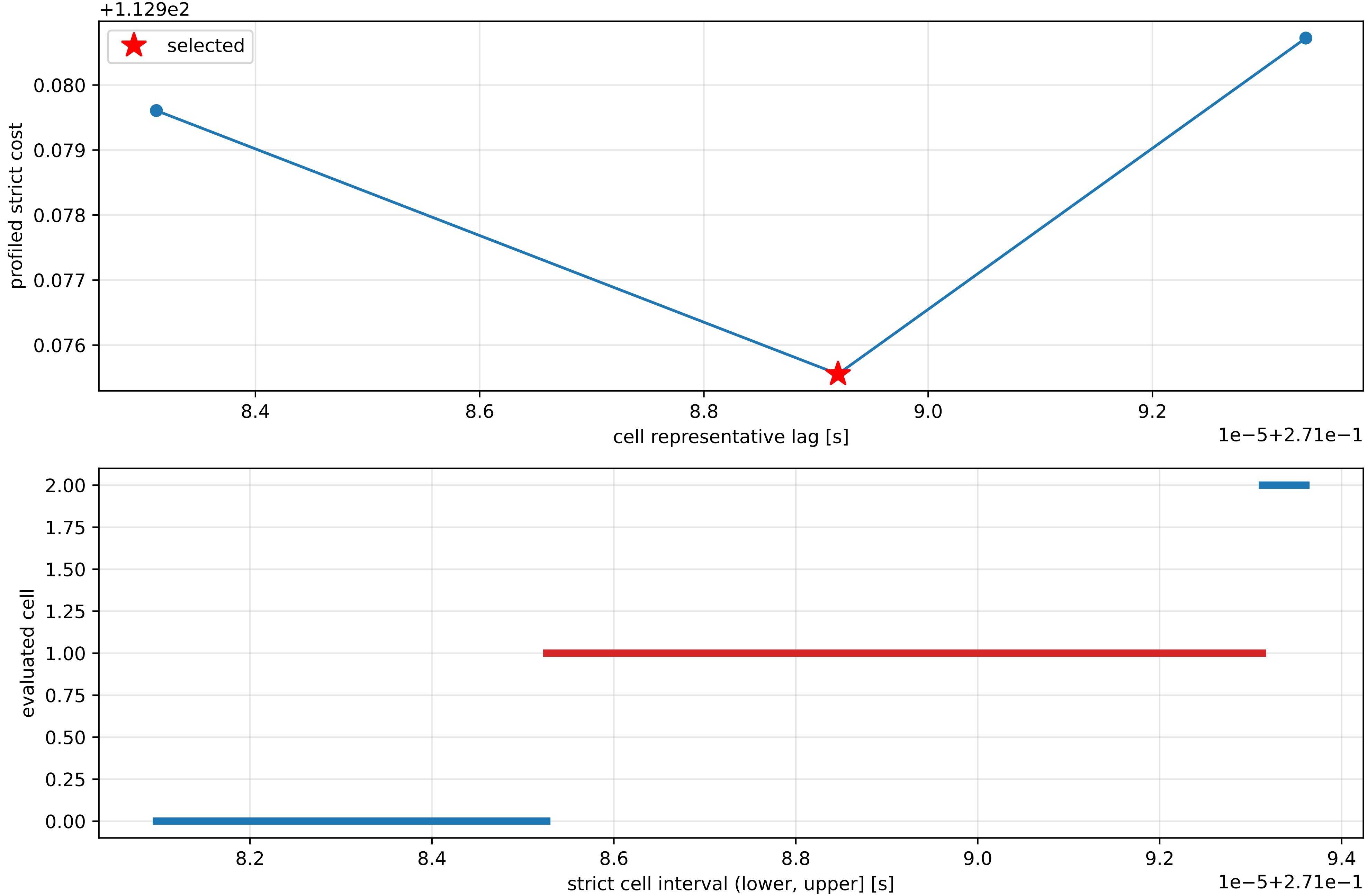


inertia_over_mass_xy_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]
time [s]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
1.0
0.5
0.0
−0.5
−1.0
0.05
0.00
−0.05
−0.10
4
3
2
1
0
−1
−2
−0.4
−0.6
−0.8
−1.0
9.8
9.6
9.4
9.2
9.0
8.8
8.6
0
1
2
3
4



inertia_over_mass_xy_nominal: smooth-to-strict continuation
rotor lag [s]
0.27115
0.27110
0.27105
0.27100
0.27095
objective
113.12
113.10
113.08
113.06
113.04
113.02
113.00
112.98
smooth
strict at same point
convergence error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

inertia_over_mass_xy_nominal: exact strict-ZOH lag cells

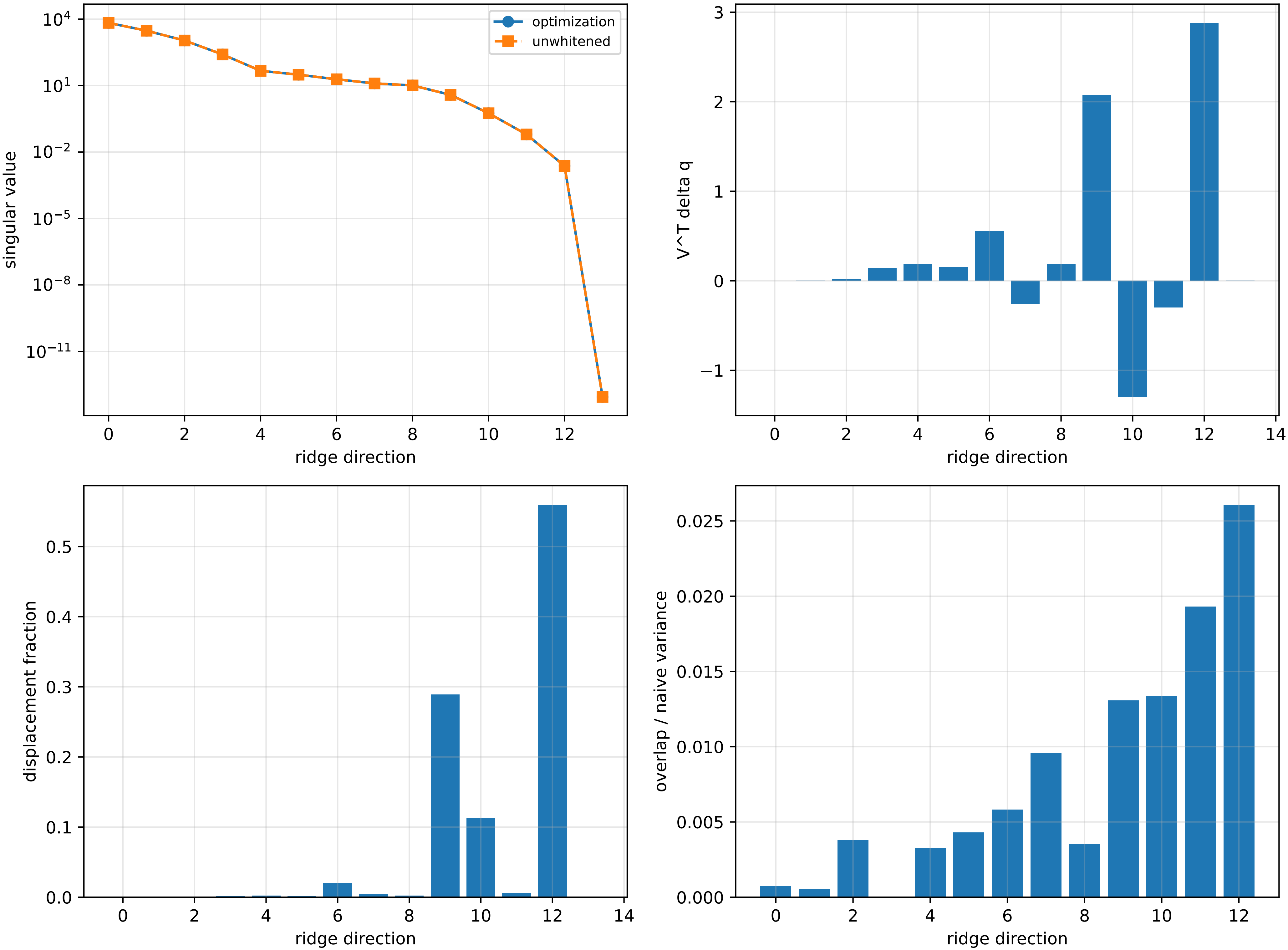


selected (0.271085262, 0.27109313] s; final smooth 0.271089137 s; data boundary=False

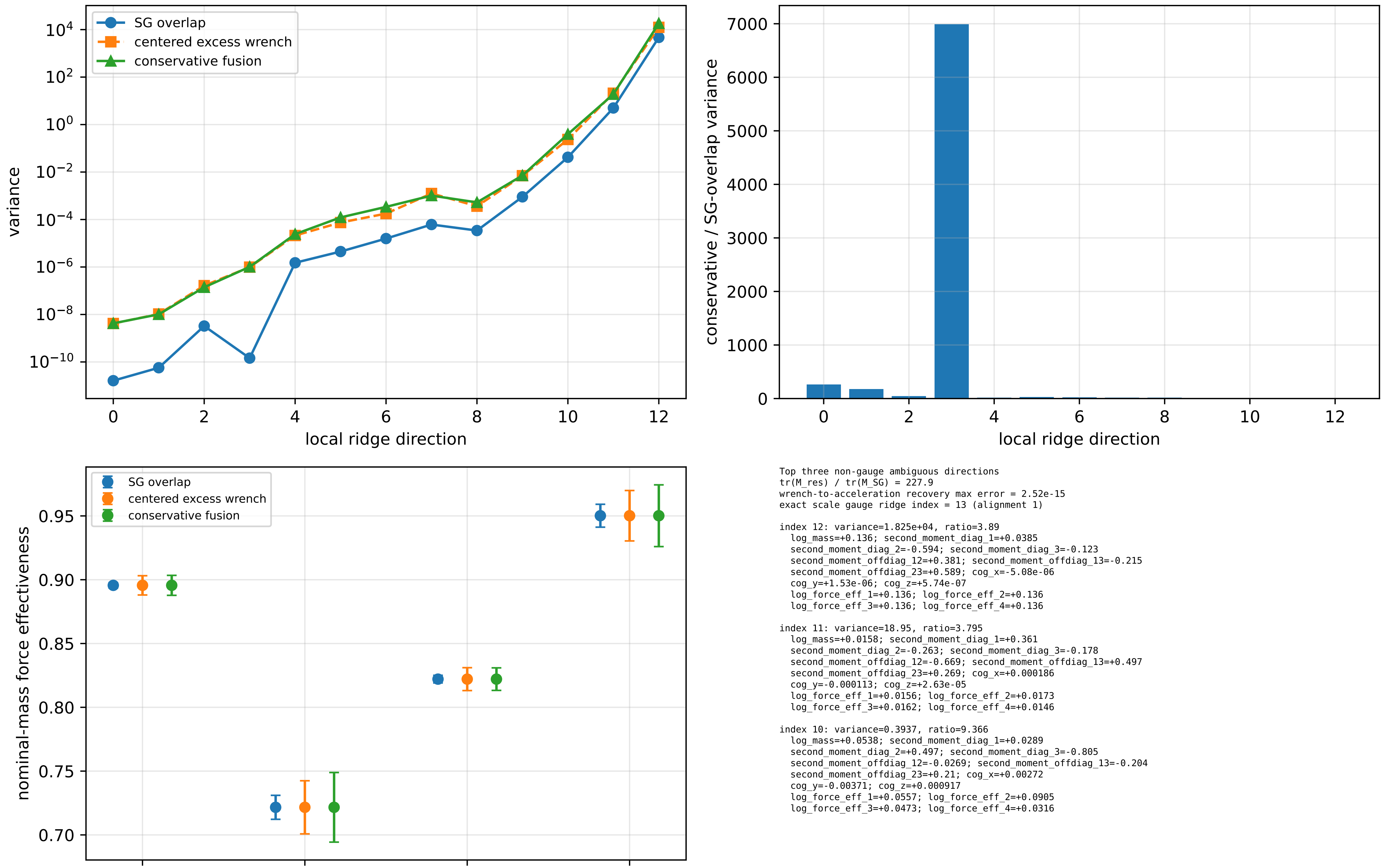
inertia_over_mass_xy_nominal: parameter estimate

```
Case: inertia_over_mass_xy_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.56156736
inertia [kg m^2]:
[[ 1.94347187e-01 -7.92001128e-07  5.30941100e-03]
 [-7.92001128e-07  9.00645194e-02 -3.09859745e-02]
 [ 5.30941100e-03 -3.09859745e-02  2.73247926e-01]]
CoG body [m]: [-0.02442724  0.01232677  0.00919404]
force effectiveness: [0.97554953 0.78605355 0.89549146 1.03499159]

scale-free J/m [m^2]:
[[ 7.58704182e-02 -3.09186142e-07  2.07271965e-03]
 [-3.09186142e-07  3.51599263e-02 -1.20964902e-02]
 [ 2.07271965e-03 -1.20964902e-02  1.06672161e-01]]
scale-free f/m: [0.38084087 0.30686429 0.34958732 0.40404621]

rotor lag cell: (0.271085262, 0.27109313] s
representative: 0.271089196 s
```

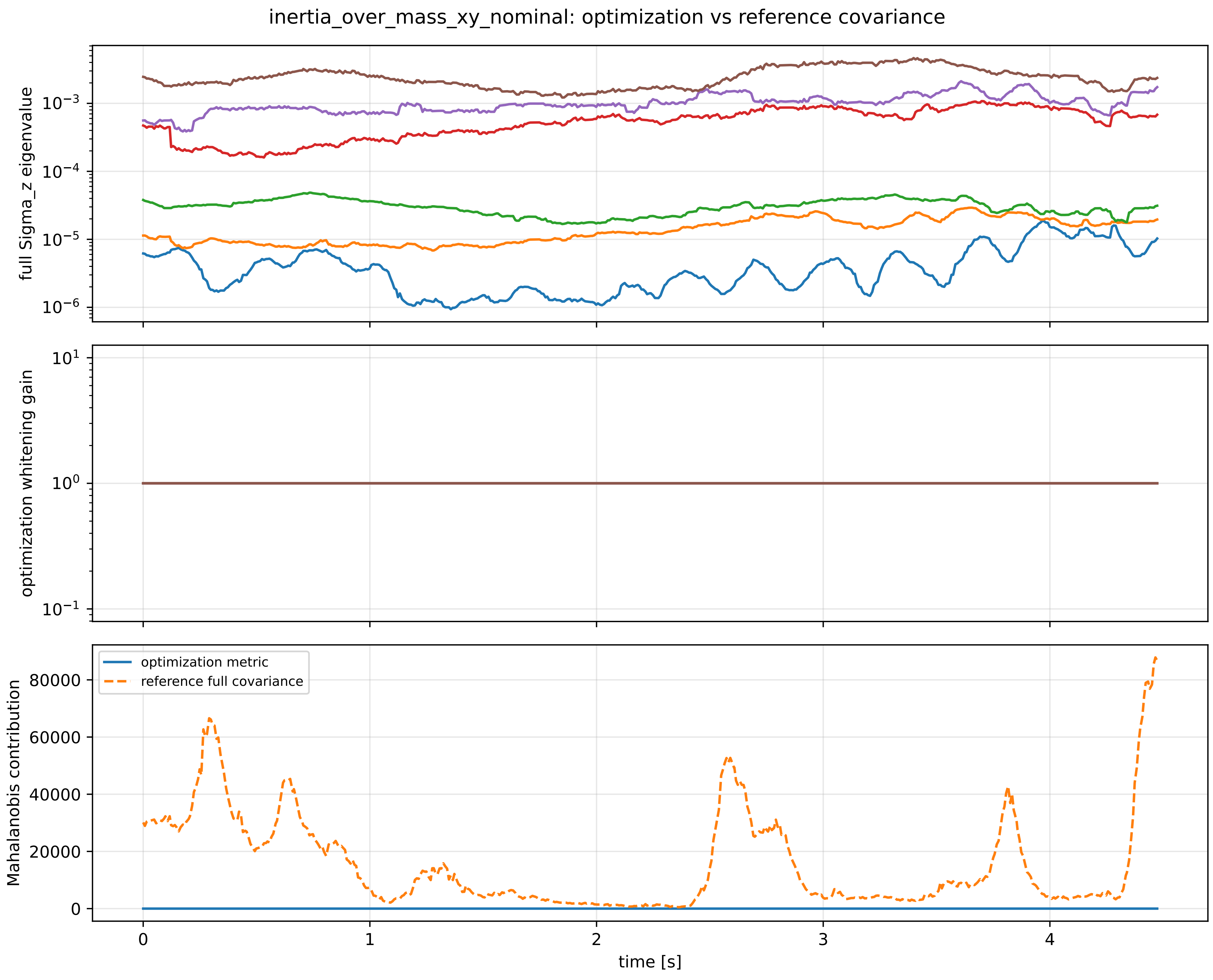
inertia_over_mass_xy_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.45e-13)



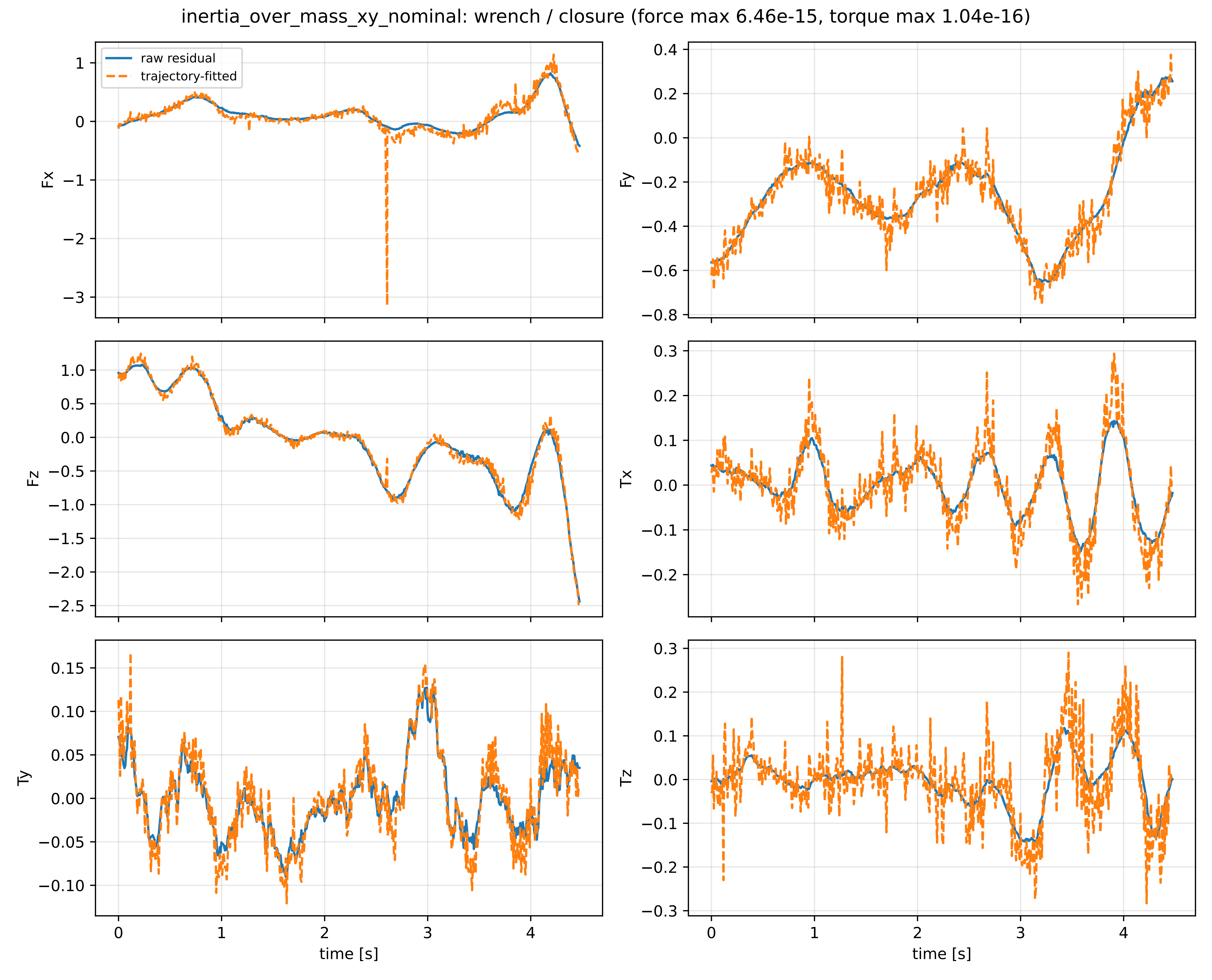
inertia_over_mass_xy_nominal: Conservative fusion uncertainty



The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.



inertia_over_mass_xy_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]


inertia_over_mass_xy_nominal: wrench / closure (force max 6.46e-15, torque max 1.04e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

inertia_over_mass_xy_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_inertia_over_mass_xy_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma
SHA256: 0e9e1cf84b655f89c02f04503cdbe3c8cb0feb8b35995087c0bc407814199394

data objective:  112.975554024
prior objective: 6.0479626503e-08
total objective: 112.975554085

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: inertia_over_mass_xy_nominal (inertia_over_mass_m2)
  components: xy
  target: [-3.09533934e-07]
  std:    [1.e-06]
  final:  [-3.09186142e-07]
  error:  [3.47791968e-10]
  standardized residual: [0.00034779]
  factor objective: 6.0479626503e-08
```